

TSING OUYANG

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EDUCATION

University of Washington, Seattle, WA Sept 2025 - Dec 2026
Master of Science in Computational Finance and Risk Management, GPA: 3.83/4.0

University of Waterloo, Waterloo, ON Sept 2020 - Apr 2025
Bachelor of Mathematics (Hons.) in Financial Analysis and Risk Management; Statistics, GPA: 3.71/4.0

FRM Part I Exam, Passed May 2023

WORK EXPERIENCE

Healthcare of Ontario Pension Plan, Performance System and Analytics Intern, Toronto, ON May 2026 – Present

- Developed and automated daily data quality audits using SQL and Python to ensure the integrity of underlying valuation models, directly supporting the accuracy of accounting and reporting calculations
- Collaborated with IT and Investment Management Reporting teams to architect data pipelines for derivative valuations, enhancing the calculation of exposure metrics and streamlining internal risk reporting processes

Athene, Actuarial Analyst Co-op, Toronto, ON Sept 2024 – Apr 2025

- Engineered a stochastic projection model using Pandas/NumPy to perform risk-neutral valuations and stress testing across 1,000+ market scenarios, quantifying Best Estimate Liabilities (BEL) to support capital adequacy assessments
- Architected an automated Python framework to calculate daily derivative collateral requirements and monitor CCR, improving calculation speed by 40% and ensuring real-time exposure management
- Executed rigorous backtesting and OLS/GLM regression analysis using SciPy to validate credit default and actuarial assumptions, achieving a 50% reduction in model variance and refining liability reserves
- Facilitated a \$13M capital allocation review by performing sensitivity analyses to identify key risk drivers, translating quantitative insights from EBS reconciliations into strategic capital adjustments

Aon, Pension Associate Intern, Toronto, ON Sept 2023 – Dec 2023

- Utilized actuarial life-contingency models to execute retirement benefit valuations and survivor entitlement projections, managing \$1M+ in disbursements with 100% data integrity
- Developed VBA automation frameworks for liability settlements and account reconciliations (COLA/PA), mitigating model risk and reducing manual processing errors by 20% across 1,000+ accounts

Liberty Utilities, FP&A Analyst Co-op, Oakville, ON Jan 2023 – Apr 2023

- Streamlined a 20GB database using SQL scripts; architected an ETL Pipeline to transform raw transactional records into structured inputs for enterprise-level forecasting models
- Conducted Linear Regression and variance analysis on utility datasets in SQL, refined cost drivers to improve budget projection accuracy and increase model R^2

PROJECTS

Bitcoin Price Predictive Trading Systems (N-BEATS) Jan 2026 – Mar 2026

- Implemented a deep-learning TensorFlow/Keras N-BEATS model to decompose non-linear cryptocurrency volatility into interpretable trend and seasonality signals through residual-based backcasting
- Engineered high-frequency data pipelines using OpenBB for 729 days of hourly data, achieving a 0.9989 R^2 and 0.0022 MAE through rigorous hyperparameter tuning and Min-Max normalization

TECHNICAL SKILLS

- *Programming*: Python (NumPy, Pandas, SciPy, TensorFlow), R (Tidyverse), Excel/VBA, SQL, C, MATLAB
- *Quantitative*: Stochastic Calculus, Black-Scholes, Monte Carlo, Martingale Pricing, Time Series (GARCH/ARIMA)
- *Tools*: Bloomberg Terminal, RESTful APIs, Snowflake, JSON, Tableau, Power BI, Git, Power Query, SAP, NetSuite